

Classifying Signals on Irregular Domains via Convolutional Cluster Pooling

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1. Motivations

We are surrounded by data lying on an underlying non-euclidean structure (e.g. 3D skeleton data, social networks and chemical compounds). Graph Convolutional Networks (GCNs) [1, 2, 3] provide a comprehensive and solid framework for *vertex classification* in such data domains, because they model directly the topological structures through edge weights. Differently, we focus on *graph signal classification*.

Our approach, built by stacking multiple **Convolutional Cluster Pooling (CCP)** layers, provides:

- a **hierarchical framework** for supervised learning in homogeneous graph contexts.
- a spatial formulation for graph filtering which, as for CNNs, exploits **weight sharing** and produces a pooled graph signal.

3. Hierarchical Soft Clustering

A cascade of M soft-partitions, described by an ordered sequence of assignment matrices $K^{(1)}, K^{(2)}, \dots, K^{(M)}$, forms a soft dendrogram for the original graph $\mathcal{A}$. The problem of obtaining a good dendrogram is formalised as follows:

$$\max_{K^{(i)}_{i=1,\dots,M}} \mathcal{L}_{\mathcal{K}} = \frac{1}{2} \sum_{m=1}^M \sum_{k=1}^{|\mathcal{K}_m|} \frac{\text{Cohesion}(K_k^{(m)})}{\text{Vol}(K_k^{(m)})}$$

subject to $\sum_{k=1}^{|\mathcal{K}_m|} K_{i,k}^{(m)} = 1 \quad i=1,2,\dots,|\mathcal{K}_{m-1}| \quad m=1,2,\dots,M.$

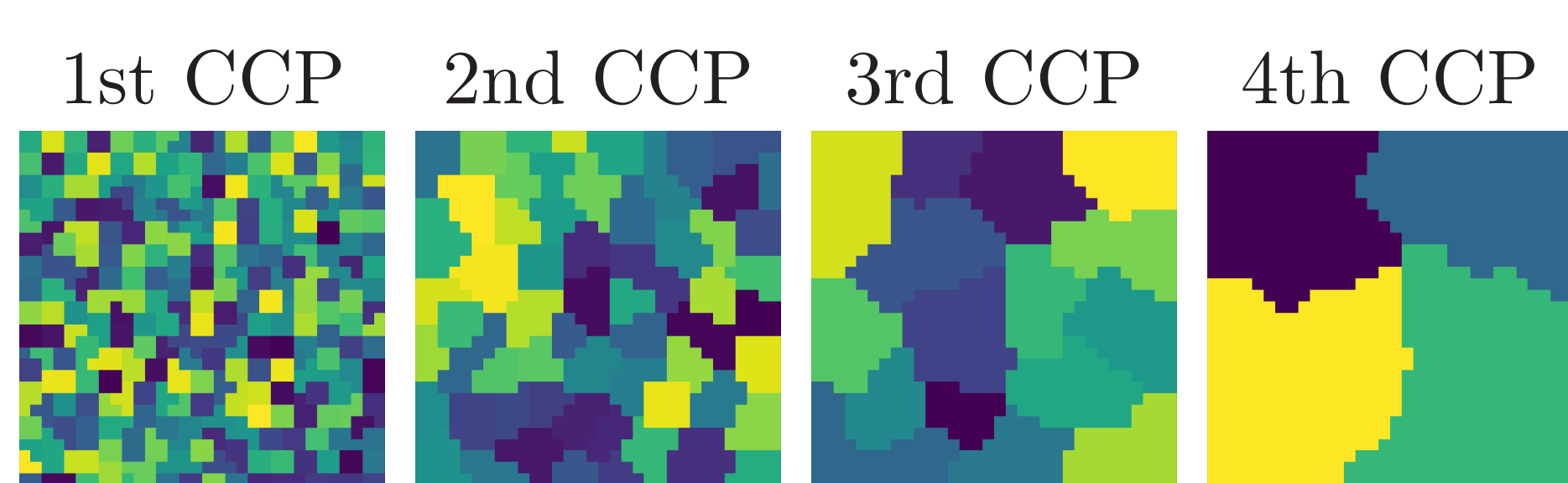
This way, we require intermediate clusters with maximal cohesion and minimum size. $\mathcal{L}_{\mathcal{K}}$ is paired to the classification loss $\mathcal{L}_0$, delivering the regularised objective $\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_{\mathcal{K}}$. This way, the supervision signal may provide information to the process of clusters formation.

6. GCN Baselines

Comparison w.r.t other graph coarsening and filtering approaches.

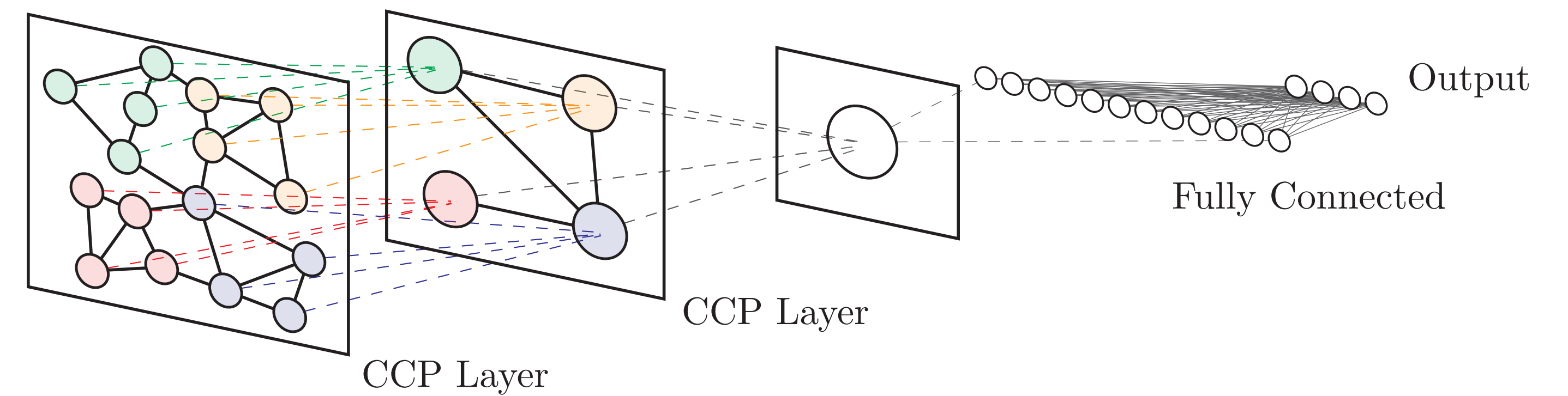
Filter	Coarsen	CIFAR-10	NTU-CS
Chebyshev [1]	Grclus	78.15	74.85
GCN [2]	Grclus	67.01	62.00
GAT [3]	Grclus	72.82	59.48
CCP	CCP	84.4	80.1

8. Learned Receptive Fields (CIFAR-10)



2. Our Proposal

Multiple applications of the CCP layer lead to multi-scale representations of the graph.

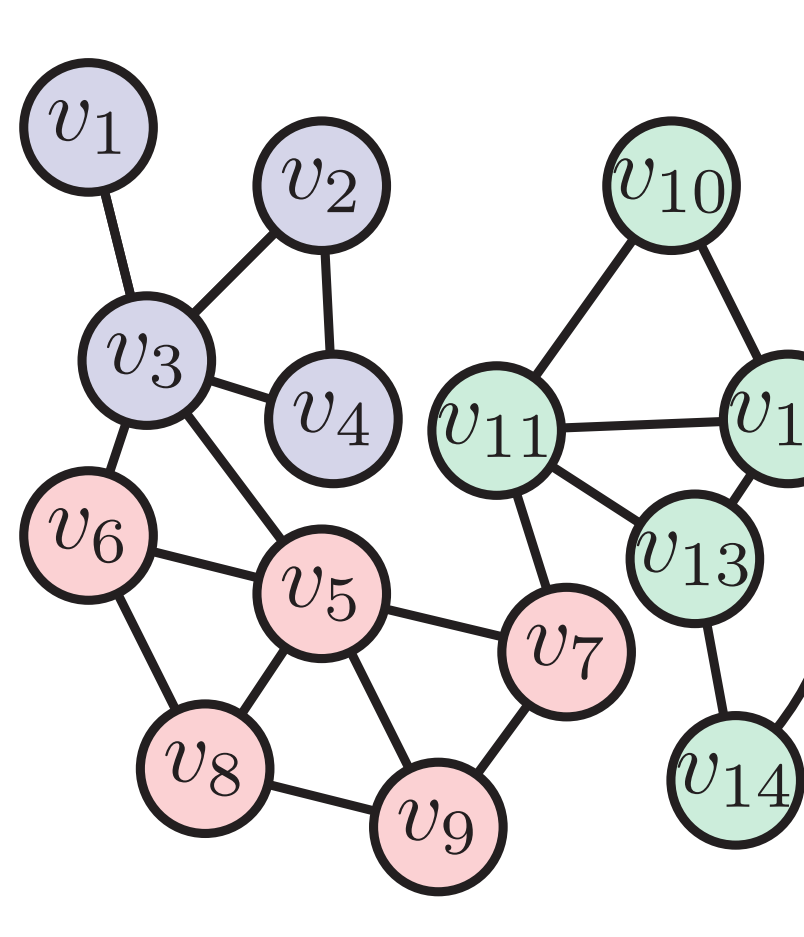


- *Firstly*, CCP performs a **clustering** operation on the input graph, resulting in a coarser output graph, whose affinity matrix reflects relationships among clusters regressed at training time.
- *Secondly*, the layer selects for each cluster a fixed number of candidate nodes for the **aggregation phase**, and sorts them depending on a **centrality-based rank** within the cluster.

4. Neighbourhood Selection

For each cluster we select as candidate set $\mathcal{N}_k^{(m+1)}$ for the filtering stage the set containing the most L representative nodes.

We consider a node more **central** if: i) it has a high membership value for the cluster under consideration; ii) a large part of its direct neighbours nodes share the same cluster in the graph.

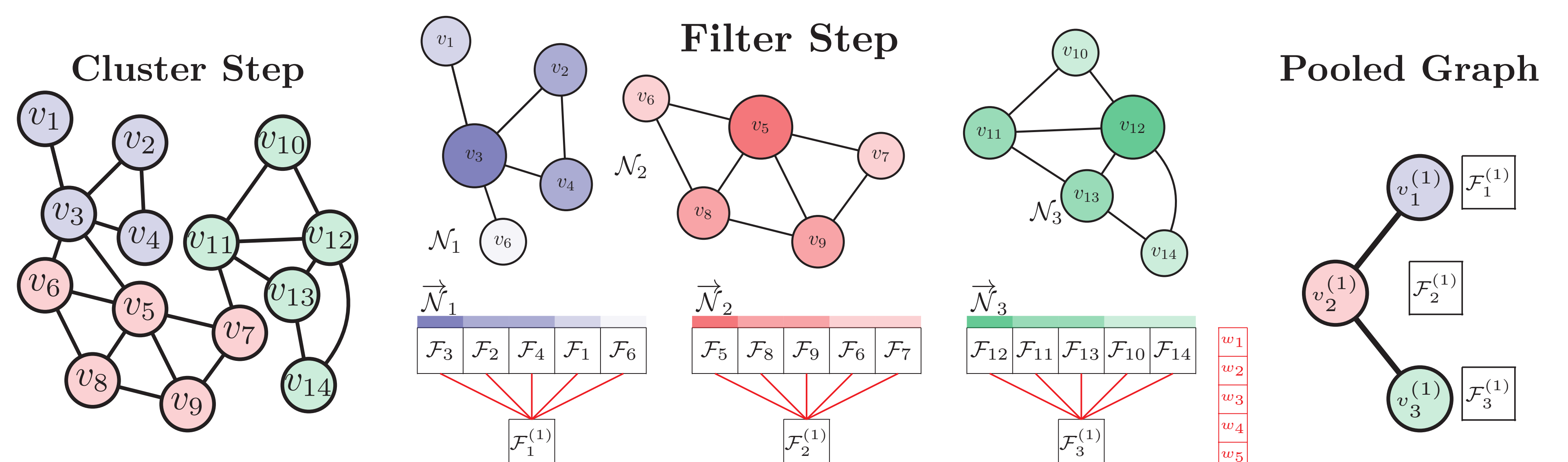


$$\text{Rank}(\mathcal{V}_i^{(m)} \rightarrow \mathcal{K}_k^{(m+1)}) = (1 + K_{i,k}^{(m+1)}) \sum_{j \neq i}^{|\mathcal{K}_m|} \mathcal{A}_{i,j}^{(m)} K_{j,k}^{(m+1)}$$

$\vec{\mathcal{N}}_1$		$\vec{\mathcal{N}}_2$		$\vec{\mathcal{N}}_3$	
$\mathcal{V}_i$	Rank	$\mathcal{V}_i$	Rank	$\mathcal{V}_i$	Rank
v_3	6	v_5	8	v_{12}	8
v_2	4	v_8	6	v_{11}	6
v_4	4	v_9	6	v_{13}	6
v_1	2	v_6	4	v_{10}	4
v_6	1	v_7	4	v_{14}	4

5. Convolutional Cluster Pooling (CCP)

The m -th CCP layer takes in input an affinity matrix $\mathcal{A}^{(m)}$ and a multi-dimensional $\mathcal{F}^{(m)} \in \mathbb{R}^{|\mathcal{K}_m| \times d_{IN}}$ signal defined on the vertex set.



CCP yields:

- $\mathcal{A}^{(m+1)}$: a new reduced affinity matrix
 - $\mathcal{F}^{(m+1)} \in \mathbb{R}^{|\mathcal{K}_{m+1}| \times d_{OUT}}$: a pooled signal
- where $W \in \mathbb{R}^{L \times d_{IN} \times d_{OUT}}$ and $b \in \mathbb{R}^{d_{OUT}}$ are parameters of our CCP layer.

$$\mathcal{F}_{k,j}^{(m+1)} = \sum_{i=1}^{d_{IN}} \sum_{l=1}^L W_{l,i,j} (\sigma_{k,l} \cdot \vec{\mathcal{N}}_k^{(m+1)}(l, i)) + b_j$$

7. Classification Results

Action recognition NTU RGB+D		Image classification CIFAR-10		Text categorisation 20NEWS	
Method	CS Acc.	Method	Acc.	Method	Acc.
P-LSTM	62.9	Graph-CNNs	68.3	Linear SVM	65.9
TGCNN	71.4	FC	78.6	FC2500-FC500	65.8
Deep STGC _K	74.9	CCP	84.4	Softmax	66.3
C-CNN	79.6	Stochastic Pooling	84.9	Chebyshev - GC32	68.3
CCP	80.1	ResNet	93.6	CCP	70.1

9. References

- [1] M. Defferrard, X. Bresson, and P. Vandergheynst. Convolutional neural networks on graphs with fast localized spectral filtering. In *NIPS*, 2016.
- [2] T. N. Kipf and M. Welling. Semi-supervised classification with graph convolutional networks. In *ICLR*, 2017.
- [3] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio. Graph Attention Networks. *ICLR*, 2018.